

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 15B.2

REDOX REACTIONS OF CARBONYL COMPOUNDS

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: ORGANIC CHEMISTRY, CARBONYLS AND CHIRALITY
SUB-TOPIC:	15B.2: Redox Reactions of Carbonyl Compounds
TOTAL QUESTIONS:	2 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	6 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 15B.2

Q1. 1. (a) Describe tests to distinguish between propanal and propanone using Tollens reagent and Fehling solution.

[4 Marks]**WCH14/01/Jun21/Q6(a)**

Q2. (b) State the reducing agent and conditions used to reduce propanone to propan-2-ol.

[2 Marks]**WCH14/01/Jun21/Q6(b)**

SECTION B: WORKED MARK SCHEME — 15B.2

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) Describe tests to distinguish between propanal and propanone using Tollens reagent and Fehling solution. 1. Tollens reagent $[\text{Ag}(\text{NH}_3)_2]^+$: propanal produces a silver mirror / grey ppt; propanone shows no reaction / stays colourless. 2. Fehling solution Cu^{2+} : propanal produces a brick-red ppt of Cu_2O ; propanone stays blue.	[4]
Q2	Q2: (b) State the reducing agent and conditions used to reduce propanone to propan-2-ol. 1. Sodium tetrahydridoborate(III), NaBH_4 (or LiAlH_4 in dry ether). 2. Aqueous ethanol / water solvent.	[2]

SECTION C: EMPIRICAL EXAMINER FAQs & COMMON TRAPS — 15B.2**FAQ #1: Why don't ketones react with Tollens or Fehling reagents?****Category: Oxidation
Reactions****COMMON EXAMINER TRAP / MISTAKE:**

Forgetting that oxidation of ketones requires breaking C-C bonds.

MODEL ANSWER / EXAMINER EXPECTATION:

Aldehydes have a C-H bond on the carbonyl carbon that is easily oxidized to C-OH (carboxylic acid). Ketones lack this H atom, so oxidation would require breaking strong C-C bonds.